(a) Run procedure

$$q(s) = q_{\text{init}} + (q_{\text{ZERO}} - q_{\text{init}})(3s^2 - 2s^3),\ s = t/T,\ 0 \le s \le 1$$

CubicTraj interpolation from the power-on pose to ZERO; joint velocity is zero at t = 0 and t = T.

1 Robot on gantry
suspended, feet unloaded

2 Power on, go to ZERO
cubic trajectory, see above

3 Lower to ground
feet loaded, gantry slack

4 Unlock, select task
gamepad, LT + RT

5 Task running
control software, see (b)

6 Power off
return to ZERO first, hoist up

photo
robot suspended on gantry

photo
robot at ZERO pose

photo
robot standing on ground

photo
operator with gamepad

next run

(b) Control software during step 5

Gamepad
command publisher

commands

LCM bus
UDP multicast

commands

Real-time control loop, 500 Hz

- Safety guards
- Task scheduling
  tasks: zero, walk, walk_amp, hold
- RL policy inference, 50 Hz
- Telemetry and logging, 50 Hz

backend interface

Real-robot backend

fieldbus
500 Hz

Robot hardware
motors, sensors

Simulation backend

physics step

MuJoCo model
physics simulation

Web panel
bridge service, read-only

telemetry

telemetry

process supervision: start / kill